

Question: 1

Given the network 192.168.10.0/25, calculate the subnet mask, network address, broadcast address, and the range of usable IP addresses?

Answer: **1. Subnet Mask**

- **Calculation:** A /25 mask means 25 bits are turned on (11111111.11111111.11111111.10000000).
- **Value: 255.255.255.128**

2. Network Address

- **Value: 192.168.10.0** (The very first address of the block, representing the subnet identity).

3. Usable IP Address Range

- **Calculation:** The host bits range from binary 0000001 to 1111110 in the last octet (decimal 1 to 126).
- **Value: 192.168.10.1 to 192.168.10.126**

4. Broadcast Address

- **Calculation:** The very last address of the block where all host bits are turned on (decimal 127).
- **Value: 192.168.10.127**

Question: 2

Your company wants to split the 10.0.0.0/24 network into four equal subnets for different departments (HR, Sales, Tech, Admin). List the network address, broadcast address, and usable IP range for each subnet?

Answer: **1. Subnet 1: HR Department (10.0.0.0/26)**

- **Network Address: 10.0.0.0**
- **Usable IP Range: 10.0.0.1 to 10.0.0.62**
- **Broadcast Address: 10.0.0.63**

2. Subnet 2: Sales Department (10.0.0.64/26)

- Network Address: 10.0.0.64
- Usable IP Range: 10.0.0.65 to 10.0.0.126
- Broadcast Address: 10.0.0.127

3. Subnet 3: Tech Department (10.0.0.128/26)

- Network Address: 10.0.0.128
- Usable IP Range: 10.0.0.129 to 10.0.0.190
- Broadcast Address: 10.0.0.191

4. Subnet 4: Admin Department (10.0.0.192/26)

- Network Address: 10.0.0.192
- Usable IP Range: 10.0.0.193 to 10.0.0.254
- Broadcast Address: 10.0.0.255

Question: 3

A startup needs 60 usable IP addresses for its team. What is the smallest CIDR subnet you can assign, and what are the network address, broadcast address, and usable IP range for this subnet?

Answer: 1. Smallest CIDR Subnet

- **CIDR Notation:** /26 (Subnet Mask: 255.255.255.192)
- **Total IPs:** 64 total addresses ($(2^6 = 64)$).
- **Usable IPs:** 62 addresses ($(64 - 2 = 62)$), which perfectly covers the 60 required hosts.

2. Network Details (Assuming a standard private base range)

- Network Address: 192.168.1.0
- Usable IP Range: 192.168.1.1 to 192.168.1.62
- Broadcast Address: 192.168.1.63

Question: 4

For the public WiFi at a Zomato restaurant, you are given the IP block 172.16.5.0/26. Calculate the subnet mask, network address, broadcast address, and the first and last usable IPs.

Hint: /26 gives you 64 addresses per subnet.?

Answer: **1. Subnet Mask**

- **Value: 255.255.255.192** (Calculated from 26 network bits: 11111111.11111111.11111111.11000000).

2. Network Address

- **Value: 172.16.5.0** (The base identifier for this public WiFi subnet).

3. Usable IP Address Scope

- **First Usable IP: 172.16.5.1** (Typically assigned to the router/default gateway).
- **Last Usable IP: 172.16.5.62** (The final available slot for a customer device).

4. Broadcast Address

- **Value: 172.16.5.63** (The final address in the 64-block chunk, reserved for routing broadcast data packets).

Question: 5

You receive the following scenario: Flipkart's warehouse needs three subnets supporting at least 30 devices each. Using CIDR notation, assign suitable subnets from the 192.168.1.0/24 range, and specify the network address, broadcast address, and usable IP range for each subnet?

Answer: **1. Subnet 1: Inventory & Scanner Devices (192.168.1.0/27)**

- **Network Address: 192.168.1.0**
- **Usable IP Range: 192.168.1.1 to 192.168.1.30**
- **Broadcast Address: 192.168.1.31**

2. Subnet 2: Automated Conveyor Systems (192.168.1.32/27)

- **Network Address: 192.168.1.32**
- **Usable IP Range: 192.168.1.33 to 192.168.1.62**

- **Broadcast Address: 192.168.1.63**

3. Subnet 3: Admin & Shipping Logistics (192.168.1.64/27)

- **Network Address: 192.168.1.64**
- **Usable IP Range: 192.168.1.65 to 192.168.1.94**
- **Broadcast Address: 192.168.1.95**